Lab 6: Instrumental Variables

PHS2000B, Spring 2026

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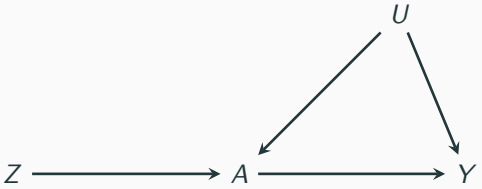
Overview

1. **Instrumental Variables**
2. **Assumptions for IV**
3. **Different types of individuals to consider in IV**
4. **IV Excersize**
5. **Situating ourselves as we transition from epidemiologic to econometric methods**

Instrumental Variables

Instrumental Variables

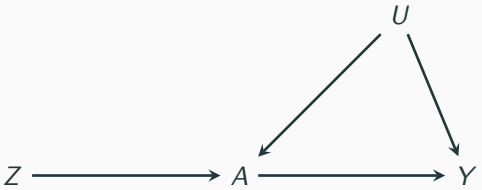
- With IV estimation, we can bypass worrying about unmeasured factors that confound the relationship between treatment and outcome.
- This can help in both experimental studies with imperfect adherence and in observation studies where people select into treatment.
- Let's start with an experimental study for a new drug, where treatment assignment, Z, is randomly assigned



- Z: assigned treatment (1: yes, 0: no)
- A: actually took the treatment (1: yes, 0: no)
- Y: outcome (Cardiovascular Disease)
- U: unmeasured factors (health knowledge)

Questions for you

1. Is the Z-A association a valid estimator of the causal effect of Z on A? (Is this association confounded?)
2. What about the association of Z-Y? Is this a valid estimator of the effect of Z on Y?
3. Will this effect tell us whether taking the treatment causes improvement in survival? Why / why not?



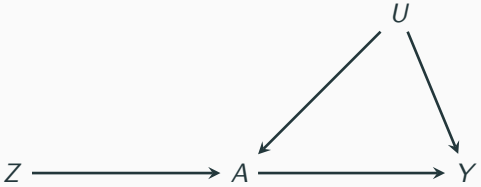
- Z: assigned treatment (1: yes, 0: no)
- A: actually took the treatment (1: yes, 0: no)
- Y: outcome (Cardiovascular Disease)
- U: unmeasured factors (health knowledge)

Intention-to-Treat vs. Per-protocol Effect

The effect of Z on Y is the **Intention-to-Treat (ITT) effect**.

The effect of A on Y that would have been observed under full adherence to assigned treatment is the **per-protocol effect.**

Based on the DAG below, can we just use the observed association between A and Y to estimate the per-protocol effect?



IV Analysis

We know:

- Effect of Z on Y (intention-to-treat effect) is unconfounded
- Effect of Z on A (adherence) is unconfounded

We can adjust the **ITT effect** by the degree of adherence to estimate the causal effect of A on Y using the standard or canonical IV estimator:

$$\text{Effect of } A \text{ on } Y = \frac{\text{Effect of } Z \text{ on } Y}{\text{Effect of } Z \text{ on } A} = \frac{ITT}{Adherence}$$

When the instrument is binary, this is $= \frac{E[Y|Z=1]-E[Y|Z=0]}{E[A|Z=1]-E[A|Z=0]}$ (Wald Estimator)

We can also estimate what is known as a **two-stage estimator** by first generating a regression for A on Z (first stage), then using predicted values of A to estimate Y on A (second stage) to reach the causal effect. This is what you will be doing in your Problem Set.

Two Stage Least Squares (2SLS)

While there is a variety of two-stage estimators, the classic is the two-stage least squares (2sls) estimator, which consists of the estimation of two OLS models:

$$\text{First Stage: } A_i = \gamma + \alpha Z_i + \mathbf{X}\beta + \mu_i$$

$$\text{Second Stage: } Y_i = \theta + \delta \hat{A}_i + \mathbf{X}\beta + \epsilon_i$$

Where the predicted values from the first model serve as the regressor in the second. Here, the estimated effect of interest is δ

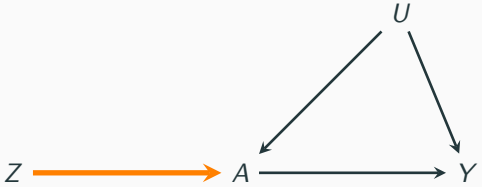
In practice, you can have multiple instruments and covariates in order to increase efficiency or if there is more than one endogenous variable. The only requirements are that there at least one instrument for every endogenous variable and every variable in the first stage is included in the second stage.

Assumptions for Instrumental Variables

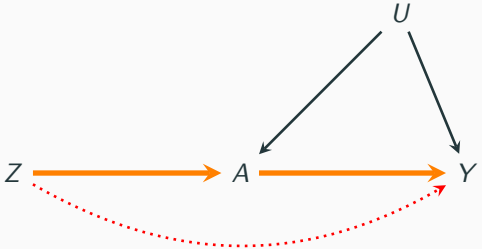
Key Assumptions for IV

Random variable Z is an instrument if it satisfies all of the following criteria:

1. **Relevance**: Z is associated with A



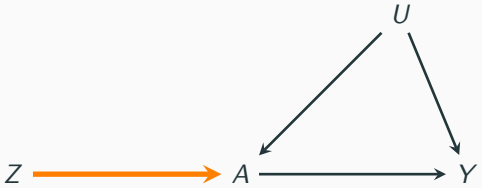
2. **Excludibility**: Z affects the outcome Y only through A



3. **Independence**:There are no common causes of the Z-Y relationship

Relevance

1. **Relevance**: Z is associated with A



For a given candidate instrument Z and exposure A, relevance is defined as:

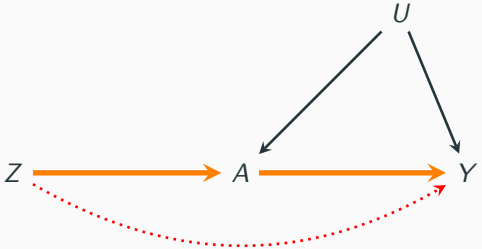
$$cov(z_i, a_i) \neq 0$$

This assumption is the **only verifiable one**, assessed during the first stage. A common rule of thumb is an F-statistic > 10, but this threshold is both arbitrary and constantly increasing in the literature. If Z is strongly associated with A, we might consider the instrument Z to be a "strong" instrument.

What would happen to our estimator $\left(\frac{\text{Effect of } Z \text{ on } Y}{\text{Effect of } Z \text{ on } A}\right)$ if the instrument was "weak"?

Excludibility

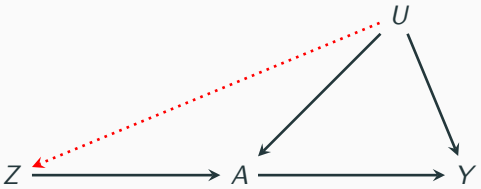
2. **Excludibility**: Z affects the outcome Y only through A



Going back to the RCT example, what would happen if blinding failed, and everyone knew their treatment assignment? Can you think of ways that simply knowing your treatment assignment might improve outcomes, separate from whether you actually take the treatment or not?

Independence

3. **Independence**:There are no common causes of the Z-Y relationship



Independence implies that there are no open back-door paths (e.g. no unmeasured confounders) from a candidate instrument Z to the outcome Y

Causal Effect Types from IV

Identifying Causal Effects

If we assume treatment effect is **homogenous**, and make the assumptions discussed above:

1. Relevance: Z is associated with the treatment A
2. Excludibility: Z affects the outcome Y only through A for all individuals in the study population
3. Independence: There are no common causes of the Z - Y relationship

IV identifies the **Average Treatment Effect (ATE)**

Identifying Causal Effects

If we assume treatment effect is **heterogenous**, we have to make an additional assumption:

1. Relevance: Z is associated with the treatment A
2. Excludibility: Z affects the outcome Y only through A for all individuals in the study population
3. Independence: There are no common causes of the Z - Y relationship
4. **Monotonicity**: The instrument affects treatment status for everyone in the same direction (i.e. no defiers)

Different kinds of individuals in the population

We can think of 4 types of individuals in a population:

Type	$A^{Z=0}$	$A^{Z=1}$
Always takers	1	1
Never takers	0	0
Compliers	0	1
Defiers	1	0

Different kinds of individuals in the population

These individual types describe counterfactual behavior, not observed behavior.

People who adhere to treatment assignments may be said to have complied with treatment assignment, but that does not automatically make them Compliers (i.e., they could be Always-takers depending on what they would have done if assigned to the control.) [1]

	$Z_i = 1$	$Z_i = 0$
$A_i = 1$	Complier/Always Taker	Defier/Always Taker
$A_i = 0$	Defier/Never Taker	Complier/Never Taker

[1] With the monotonicity assumption (plus a few more), you can estimate the proportion of the population that are compliers! This is easiest to see when the instrument and treatment are both binary.

Local Average Treatment Effect

The monotonicity assumption implies that there are no defiers. This means that the local average treatment effect will only be the effect among compliers (i.e., folks whose exposure is changed by the instrument).

LATE is the treatment effect among compliers

$$\text{LATE} = E[Y^{a=1} - Y^{a=0} | A^{z=1} = 1, A^{z=0} = 0]$$

Homogeneity: when LATE = ATE

Homogeneity in IV is the when we think that the effect among compliers would be the same as everyone else (always-takers, never-takers, and defiers). Under this assumption, we can see that the estimated LATE is equal to the ATE.

$$\text{ATE} = [\text{effect among AT}] \times \pi_{11} + [\text{effect among NT}] \times \pi_{00} + [\text{effect among D}] \times \pi_{10} + [\text{LATE}] \times \pi_{01}$$

$$\text{ATE} = [\text{LATE}] \times \pi_{11} + [\text{LATE}] \times \pi_{00} + [\text{LATE}] \times \pi_{10} + [\text{LATE}] \times \pi_{01}$$

$$\text{ATE} = [\text{LATE}] \times (\pi_{11} + \pi_{00} + \pi_{10} + \pi_{01})$$

$$\text{ATE} = [\text{LATE}] \times (1)$$

$$\text{ATE} = \text{LATE}$$

How reasonable is the homogeneity assumption?

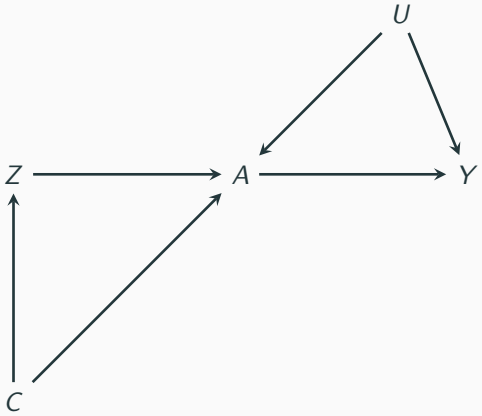
Interpreting Causal Effects

Let's imagine an RCT where participants are randomly assigned whether to take a daily aspirin. We are interested in the effects on the prevalence of cardiovascular disease (CVD).

- **ITT**: This is the effect of randomly being assigned to take aspirin on CVD
- **ATE**: This is the effect of actually taking aspirin on CVD. If we are only comfortable assuming monotonicity but not homogeneity, we can't actually estimate this.
- **LATE**: This is the effect of actually taking aspirin on CVD among the group of people who would only have taken aspirin if assigned to do so (i.e., the compliers).

Conditional exogeneity

Sometimes there is no 'clean' exogenous instrument that can be seen as truly random. In these cases, we can use a conditional instrument, where exogeneity is achieved by conditioning on observable confounders in the Z-A causal relationship.



Conditional exogeneity

While this is an attractive way to estimate causality in even more complex settings, the use of covariates does raise a few potential issues:

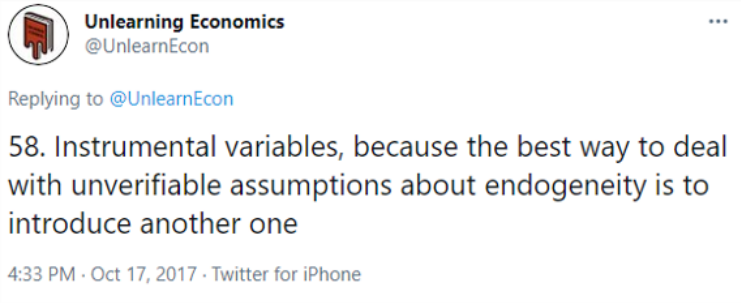
- Mechanics (i.e. how the first and second stage regressions are estimated)
- Interpretation
- Confidence in assumptions (i.e. Independence)

Do you bel-IV?

It is one thing to have randomization as an instrument where independence is likely to hold.

However, the draw of IV historically has been the ability to find an instrument that can account for a host of various confounders.

The key argument is that there is no way the instrument affects the outcome except through the treatment of interest. How plausible is this?



It's Rainin' IV!

Rainfall is often used as an instrument for different questions. For example the effect of parental income on education in agricultural settings. Or TV viewing on child development (rain causes kids to watch TV instead of playing outside).

But if rainfall is affecting various exposures of interest.... then perhaps it is not a good instrument?

IV Excersize

Does Compulsory School Attendance Affect Schooling and Earnings?

To estimate the income returns to education without confounding, the authors use the quarter of birth as an instrumental variable for schooling. The idea is that if you are born at the beginning of the year (i.e., the first quarter), then you start the school at an older age and therefore can drop out after completing less schooling than those who were born near the end of year. The authors use the two-stage least squares regression to estimate the effects of schooling on wages. We analyze the data from the **1925** cohort from this study. [2]

variable	description
`EDUC`	number of years in education
`LWKLWGE`	log weekly wage in 1980
`QOB`	quarter of birth
`first_quarter`	binary variable for whether the person was born in the first quarter

[2] Angrist, Joshua D., and Alan B. Krueger. "Does Compulsory School Attendance Affect Schooling and Earnings?" (https://doi.org/10.2307/2937954)" *The Quarterly Journal of Economics* 106, no. 4 (1991): 979-1014.

Question 1

Estimate and interpret the effect of first_quarter birth on years of education.

How would you evaluate the relevance of this instrument?

Question 1

```r
summary(iv_first_stage)
```

```
##
## Call:
## lm(formula = EDUC ~ first_quarter, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.5234  -2.4470   0.4766   1.4766   6.5530
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   11.52336    0.02482 464.262   <2e-16 ***
## first_quarter -0.07634    0.04912  -1.554     0.12
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.375 on 24832 degrees of freedom
## Multiple R-squared:  9.728e-05,  Adjusted R-squared:  5.701e-05
## F-statistic: 2.416 on 1 and 24832 DF,  p-value: 0.1201
```

Being born in the first quarter causes a 0.07634 reduction in the years of education attained.

F-statistic of 2.416 suggests a weak instrument

Question 2

Estimate the causal effect of first-quarter birth on log weekly wages

How would you interpret this estimate?

Question 2

```r
summary(iv_reduced)
```

```
##
## Call:
## lm(formula = LWKLYWGE ~ first_quarter, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.1854 -0.2368  0.0456  0.3400  3.3981
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.165647   0.004774 1082.12   <2e-16 ***
## first_quarter -0.010583   0.009446   -1.12    0.263
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6491 on 24832 degrees of freedom
## Multiple R-squared:  5.054e-05,  Adjusted R-squared:  1.027e-05
## F-statistic: 1.255 on 1 and 24832 DF,  p-value: 0.2626
```

Being born in the first quarter causes a 0.010583 reduction in the average log-weekly wage in 1980.

Question 3

Using only the estimates already calculated, how would you estimate the LATE?

Question 3

$$LATE = \frac{ITT}{First\ \ Stage} = \frac{-0.01058}{-0.07634} = 0.13859051611$$

Question 4

'Manually' conduct the 2sls estimation of the LATE and interpret

Question 4

```r
data$EDUC_predict <- predict(iv_first_stage)
iv_2sls <- lm(LWKLYWGE ~ EDUC_predict, data=data)
summary(iv_2sls)
```

```
##
## Call:
## lm(formula = LWKLYWGE ~ EDUC_predict, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.1854 -0.2368  0.0456  0.3400  3.3981
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.5683     1.4235   2.507   0.0122 *
## EDUC_predict     0.1386     0.1237   1.120   0.2626
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6491 on 24832 degrees of freedom
## Multiple R-squared:  5.054e-05,  Adjusted R-squared:  1.027e-05
## F-statistic: 1.255 on 1 and 24832 DF,  p-value: 0.2626
```

Among those induced to receive less years of education due to being born in the first quarter, an additional year of education causes a 0.1386 increase in the average log-weekly wage in 1980

Bringing it home

```r
library(ivmodel)
```

```
## Warning: package 'ivmodel' was built under R version 4.2.3
```

```r
ivmodel(Y=data$LWKLYWGE, D=data$EDUC, Z=data$first_quarter)
```

```
##
## Call:
## ivmodel(Y = data$LWKLYWGE, D = data$EDUC, Z = data$first_quarter)
## sample size: 24834
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## First Stage Regression Result:
##
## F=2.415849, df1=1, df2=24832, p-value is 0.12013
## R-squared=9.727826e-05,   Adjusted R-squared=5.701156e-05
## Residual standard error: 3.375266 on 24833 degrees of freedom
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## Coefficients of k-Class Estimators:
##
##               k Estimate Std. Error t value Pr(>|t|)
## OLS    0.000000 0.081243   0.001106  73.444   <2e-16 ***
## Fuller 0.999960 0.121822   0.096843   1.258    0.208
## TSLS   1.000000 0.138621   0.118076   1.174    0.240
## LIML   1.000000 0.138621   0.118076   1.174    0.240
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## Alternative tests for the treatment effect under H_0: beta=0.
```

One week of metrics down!

Thank you Sudipta Saha and Austin Tucker for laying the foundation for this lab and Professor Kosuke Imai for the IV example

Situating Ourselves

Culture shock: moving between epidemiology and econometrics

As we move into the econometrics section of the semester, it is easy to be overwhelmed by changes in terminology and methods.

It may help to first focus on the commonalities before the stressing over the differences

What is common?

We are still trying to identify causal effects, so counterfactual / potential outcomes thinking is still helpful!

Epidemiologists and econometricians both rely on strong assumptions. Our work is shaped by our worldviews, beliefs and explicit or implicit theories.

What is different? Big Picture

Epidemiologists

- Measurement of confounders, DAGs, and various methods of quantifying unmeasured confounding, adjustment and bounding for confounding
- Often start with the question of interest, then identify or gather data to answer it

Econometricians

- Generally skeptical that confounding can be controlled such that no unmeasured confounding remains (DAGs only recently seeing more usage)
- May start identifying possible natural/quasi-experimental opportunities, or even with identifying data, and then asking what causal questions could be answered

What is different? ENDOGENEITY

If we have a model $Y_i = \beta_i X_i + \epsilon_i$ and $cov(X, \epsilon) \neq 0$ then we have endogeneity

We can have endogeneity for several reasons:

- **Omitted variables**(omitted variable bias)
- **Measurement error in independent variable**
- **Simultaneity/reverse causality**

Endogeneity can cause our estimates to be biased and inconsistent

What is different? EXOGENEITY

Definition: a variable whose value is determined by factors outside the causal model

If we have a model $Y_i = \beta_i X_i + \epsilon_i$ and $cov(X, \epsilon) = 0$ then we have exogeniety

Where may we have seen exogeniety?

Epidemiology and Econometrics - a synthesis

- This course is giving a bird's eye view of various methods in different disciplines
- Different methods from different fields will be suitable for different questions. And the same kind of questions can be answered using different methods, depending on context and data sources
- There is increasing work across disciplines - e.g. social epidemiologists using econometric methods, econometricians using DAGs and potential outcomes framework.[3]

[3] Imbens, Guido W. 2020. "Potential Outcome and Directed Acyclic Graph Approaches to Causality: Relevance for Empirical Practice in Economics." Journal of Economic Literature, 58 (4): 1129-79. DOI: 10.1257/jel.20191597